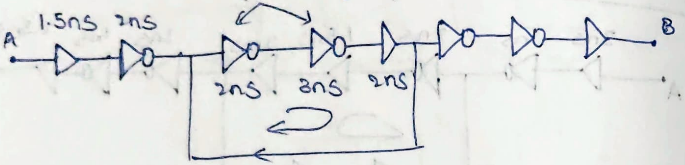
(*) If the given loop has even no. of NOT gates in loop, then it is called as Mono stable Multivibrator.

-> It'll have zero frequency, until you use any trigger.

(2 NOT gates)

(a) 

A: Freq. = 0 (Even no. of inverters)
(No oscillations)

=> DeMorgan's Theorem :-

1) $\overline{A \cdot B} = \overline{A} + \overline{B}$

2) $\overline{A+B} = \overline{A} \cdot \overline{B}$

★ $\overline{A+B+C} = \overline{A} \cdot \overline{B} \cdot \overline{C}$

★ $\overline{A \cdot B \cdot C} = \overline{A} + \overline{B} + \overline{C}$

END OF DAY 1

(*) Important Points :-

-> $A \cdot 1 = A$ -> $A \cdot A = A$ -> $A + A = A$

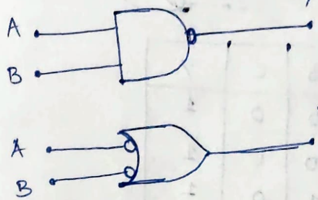
-> $A \cdot 0 = 0$ -> $A \cdot \overline{A} = 0$ -> $A + 0 = A$

-> $A + 1 = 1$ -> $A + \overline{A} = 1$

=> Universal Gates DAY-2 (01-06-26)

=> NAND Gate :- Combination of AND + NOT gates.

$Y = \overline{AB} = \overline{A} + \overline{B}$



$Y = \overline{A} + \overline{B}$

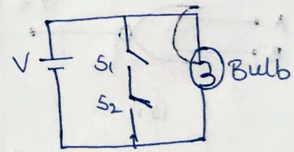
NAND = Bubbled OR gate.

Truth Table -

A	B	Y
0	1	1
1	0	1
0	0	1
1	1	0

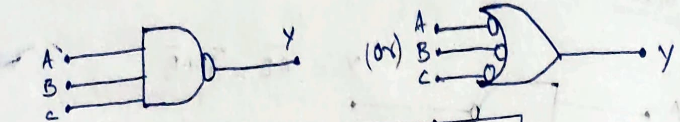
O/P is low, if both inputs are high. Else all cases are High.

(*) Switching circuit -



Truth Table -

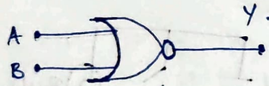
S_1	S_2	Bulb
ON	OFF	ON
OFF	ON	ON
OFF	OFF	ON
ON	ON	OFF



Truth Table -

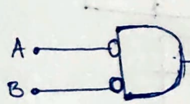
A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

⇒ <u>NOR Gate</u> :- Combination of (OR + NOT) gates



$Y = \overline{A+B} = \overline{A} \cdot \overline{B}$

(or)



$Y = \overline{A} \cdot \overline{B}$

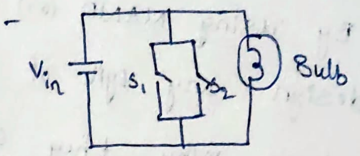
NOR = Bubbled AND gate

Truth Table -

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

⇒ If both I/p's are low, o/p will be High.

(*) Switching circuit -

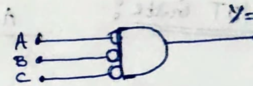


Truth Table -

S_1	S_2	Bulb
ON	OFF	OFF
OFF	ON	OFF
OFF	OFF	ON
ON	ON	OFF

(*) 3 i/p NOR Gate -

$Y = \overline{A+B+C}$



$Y = \overline{A+B+C}$

Truth Table -

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

→ WHY Universal gates :-

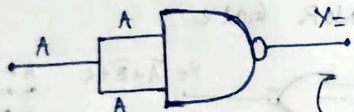
→ By using NAND (or) NOR gates, you can design any type of gate.

→ That's why they are called Universal gates.

→ Also, you can implement any gate using NAND (or) NOR gates.

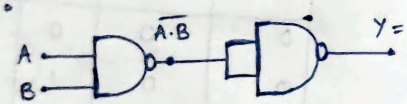
(*) NAND Gate as universal gate :-

① NOT Gate :-



$Y = \overline{A \cdot A} = \bar{A}$

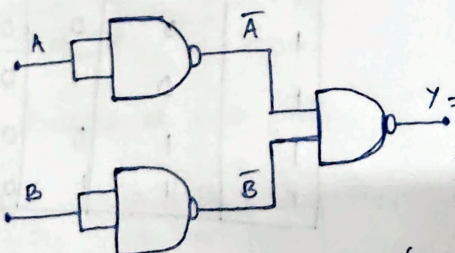
② AND Gate :-



$Y = \overline{\overline{AB} \cdot \overline{AB}}$

$= AB$.

③ OR Gate :-



$Y = \overline{\bar{A} \cdot \bar{B}}$

$= \bar{\bar{A}} + \bar{\bar{B}}$

$= A + B$.

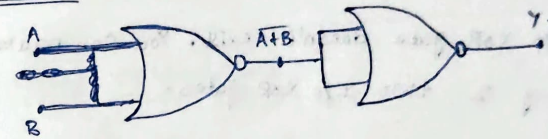
(DeMorgan's Law)

(*) NOR gate as Universal gate :-

① NOT gate :-

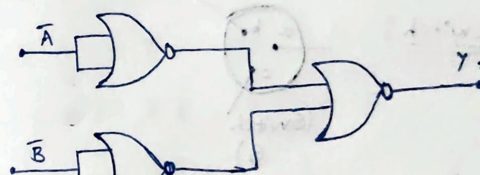
$B = \overline{A + A} = \bar{A}$.

② OR gate :-



$Y = A + B$

③ AND Gate :-



$Y = \overline{\bar{A} + \bar{B}} = A \cdot B$.

(DeMorgan's Law)

⇒ ARithmetic Gates :-

① EX_OR / XOR gates :- O/P is High when both I/P are different. O/P is Low when both I/P are same.

$Y = \bar{A}B + A\bar{B} = A \oplus B$

$\boxed{\begin{array}{c} AB = BA \\ A+B = B+A \end{array}}$

(XNOR detector) XOR

Truth Table

A	B	$\bar{A}$	$\bar{B}$	Y
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	0

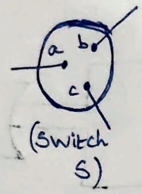
=> $0 \oplus 0 = 0$
$0 \oplus 1 = 1$
$1 \oplus 0 = 1$
$1 \oplus 1 = 0$

Note: 3 I/p XOR gate doesn't exist. You can make it using 2 two I/p XOR gates.

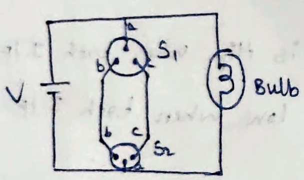
XOR gate = odd no. of 1's detector

(*) $\boxed{A \oplus \bar{A} = 1}$

(*) Two way switch:-



=> when a is connected to b, S is ON.
when a is connected to c, S is OFF.

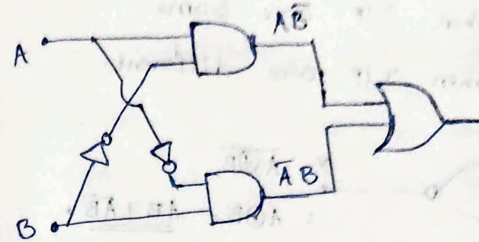


S_1	S_2	Bulb
ON	ON	OFF
ON	OFF	ON
OFF	ON	ON
OFF	OFF	OFF

=> $A \oplus B = B \oplus A$

=> $A \oplus B \oplus C = (A \oplus B) \oplus C = \underline{A \oplus (B \oplus C)}$

=> XOR gate from Basic gates:-



$Y = A\bar{B} + \bar{A}B$
$= A \oplus B$

=> Important Observations:-

① $A \oplus A = A\bar{A} + \bar{A}A = 0$

② $A \oplus 0 = A \cdot 1 + \bar{A} \cdot 0 = A$

③ $A \oplus 1 = A \cdot 0 + \bar{A} \cdot 1 = \bar{A}$

④ $A \oplus \bar{A} = A \cdot A + \bar{A} \cdot \bar{A} = A + \bar{A} = 1$

⑤ $A \oplus B = C$; $A \oplus C = ?$ => $B \oplus C = \underline{A}$
$= A \oplus (A \oplus B)$
$= 0 \oplus B = \underline{B}$.

(Q) Find

① $A \oplus A \oplus A \oplus A = (A\bar{A} + \bar{A}A) \oplus (A\bar{A} + \bar{A}A)$
$= 0 \oplus 0 = \underline{0}$.

② $A \oplus A \oplus A \oplus A \oplus A \oplus A = \underline{0}$.

③ $A \oplus A \oplus A = 0 \oplus A = \underline{A}$.

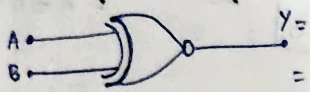
④ $A \oplus A \oplus A \oplus A \oplus A = \underline{A}$.

=> Even no. of A's = 0 => odd no. of A's = A.

⇒ EX-NOR (or) XNOR :- [XOR + NOT]

o/p is High when I/p are Same.

o/p is low when I/p are different.



$Y = \overline{A \oplus B}$

$= A \odot B = AB + \bar{A}\bar{B}$.

A	B	$\bar{A}$	$\bar{B}$	Y
0	0	1	1	1
0	1	1	0	0
1	0	0	1	0
1	1	0	0	1

→ (Equivalence detector)

⇒ Solve :-

① $1 \odot 1 \odot 1 = 1$

③ $0 \odot 0 \odot 0 = 0$

② $1 \odot 1 \odot 1 \odot 1 \odot 1 = 1$

④ $0 \odot 0 \odot 0 \odot 0 = 1$

⑤ $1 \odot 0 \odot 0 \odot 1 \odot 0 = 0$

⑥ $1 \odot 0 \odot 0 \odot 1 \odot 0 \odot 0 \odot 1 = 1$

(★) XOR Gate = odd no. of 1's detector
XNOR Gate = even no. of 0's detector

⇒ Odd no. of 0's = 0
Even no. of 0's = 1 (Short cut)

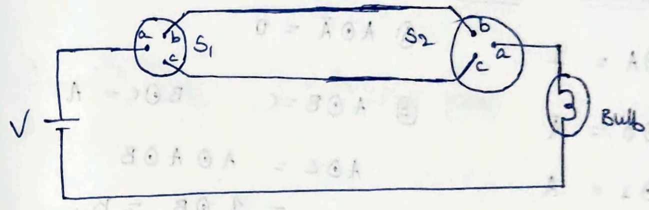
★ $A \oplus B = \overline{A \odot B}$

$A \oplus B \oplus C = A \odot B \odot C$

⇒ For even no. of I/p ⇒ XOR = $\overline{\text{XNOR}}$

odd " " " ⇒ XOR = XNOR.

(*) Switching CKT

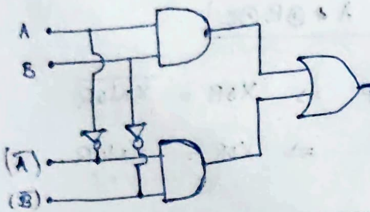


S_1	S_2	Bulb
ON	ON	ON
ON	OFF	OFF
OFF	ON	OFF
OFF	OFF	ON

⇒ $A \odot B = B \odot A$

⇒ $A \odot [B \odot C] = [A \odot B] \odot C$

⇒ XNOR using other gates :-



$Y = AB + \bar{A}\bar{B}$

⇒ Important observations :-

① $A \odot A = 1$

② $A \odot 0 = \bar{A}$

③ $A \odot 1 = A$

④ $A \odot \bar{A} = 0$

⑤ $A \odot B = C$, $B \odot C = A$

$A \odot C = A \odot A \odot B$

$= 1 \odot B = B$

(Q) Find

① $A \odot A \odot A = A$

② $A \odot A = 1$

③ $A \odot A \odot A \odot A = 1$

④ $A \odot A \odot A \odot A \odot A = A$

Even no. of A = 1

odd " " " = A

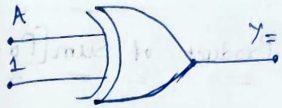
⇒ $\bar{A} \oplus B = A \odot B = \overline{A \oplus B}$

⇒ $\bar{A} \oplus \bar{B} = A \oplus B$

⇒ $\bar{A} \odot B = A \oplus B = \overline{A \odot B}$

⇒ $\bar{A} \odot \bar{B} = A \odot B$

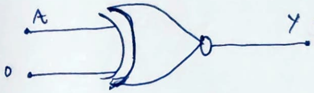
⇒ Buffer, NOT gates using XOR, XNOR :-

⇒ 

$Y = \bar{A}$ (NOT gate)

⇒ 

$Y = A$ (Buffer)

⇒ 

$Y = \bar{A}$ (NOT gate)

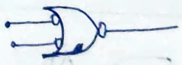
⇒ 

$Y = A$ (Buffer)

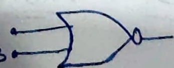
⇒ Equivalent Gates :-

①  ≈ 

NAND — OR Bubbled

②  ≈ 

AND — NOR

③ 

≈ 

NOR — AND Bubbled

④ 

(*) Sum of Product [SOP] & Product of Sum [POS] :-

① SOP :- AB + CD, AB + BC + CD

② POS :- (A+B)(C+D), (A+B)(B+C)(C+D)

⇒ Implementation using Min. no. of 2 I/p NAND (or) NOR Gates :-

⇒ Min. no. of NAND Gate :-

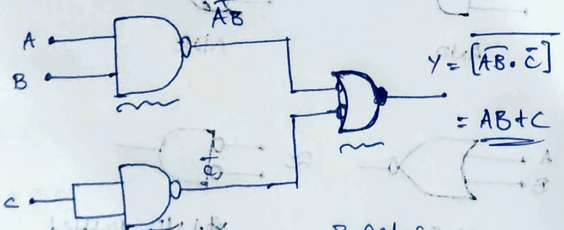
① Minimize the function. [Try Converting in SOP]

② Implement fn. using Basic gates.

③ Convert Basic gates to NAND gates.

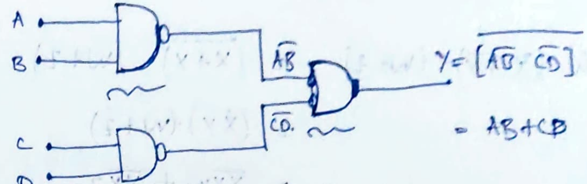
	No. of NAND gates
NOT –	1
AND –	2
OR –	3

(Q) AB+C



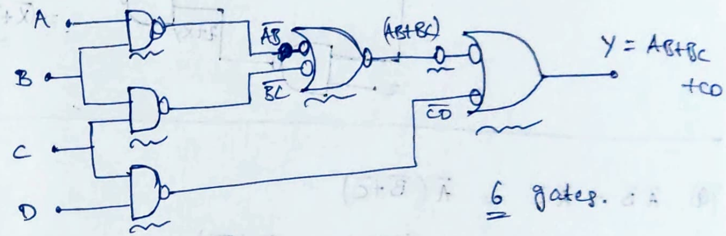
3 gates

(Q) AB+CD



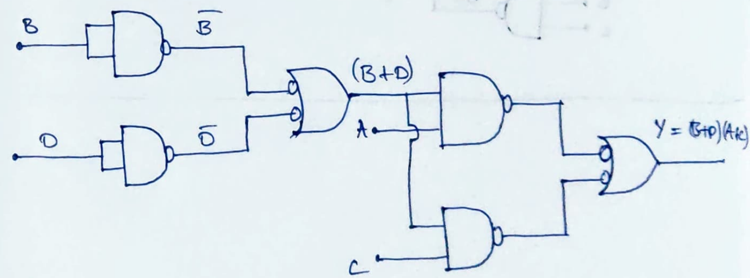
3 gates

(Q) AB + BC + CD

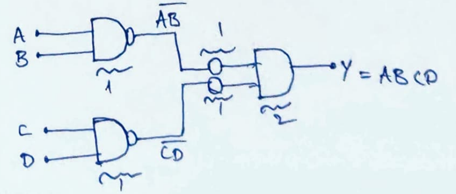


6 gates.

(Q) AB + BC + CD + DA = (A+C)B + (A+C)D
= (B+D)(A+C)



(Q) ABCD

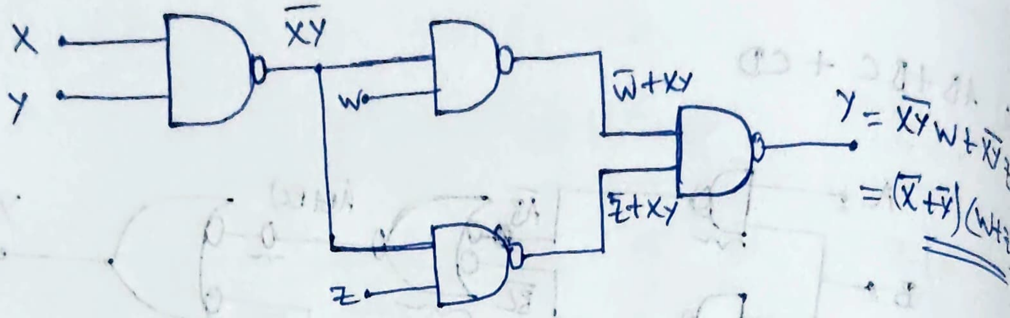


6.

(Q) $(\bar{X}+\bar{Y})\cdot(W+Z) = \overline{(X+Y)}\cdot(W+Z)$

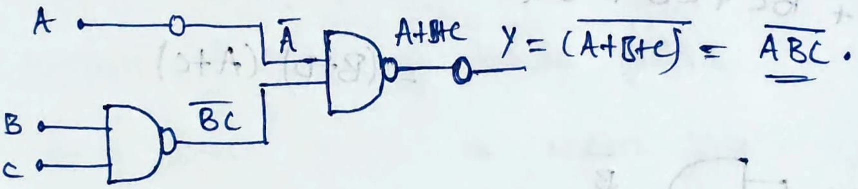
$= (\overline{XY})\cdot(W+Z)$

$= \overline{XY}W + \overline{XY}Z$



(Q) $\bar{A}\bar{B} + \bar{A}\bar{C} = \bar{A}(\bar{B}+\bar{C})$

$= \bar{A}\,(\overline{B+C}) = \bar{A}\cdot(\overline{BC})$



END OF DAY 2 //